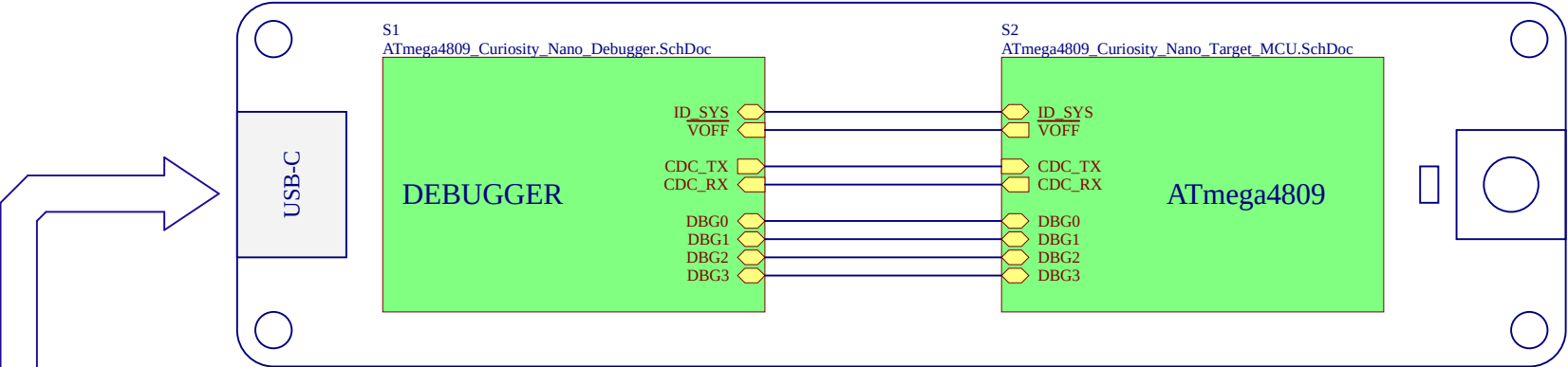
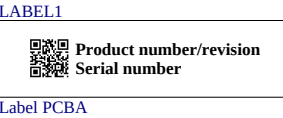
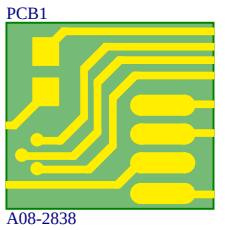
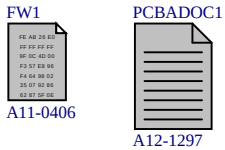
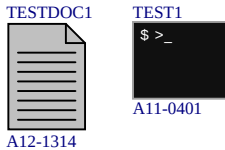
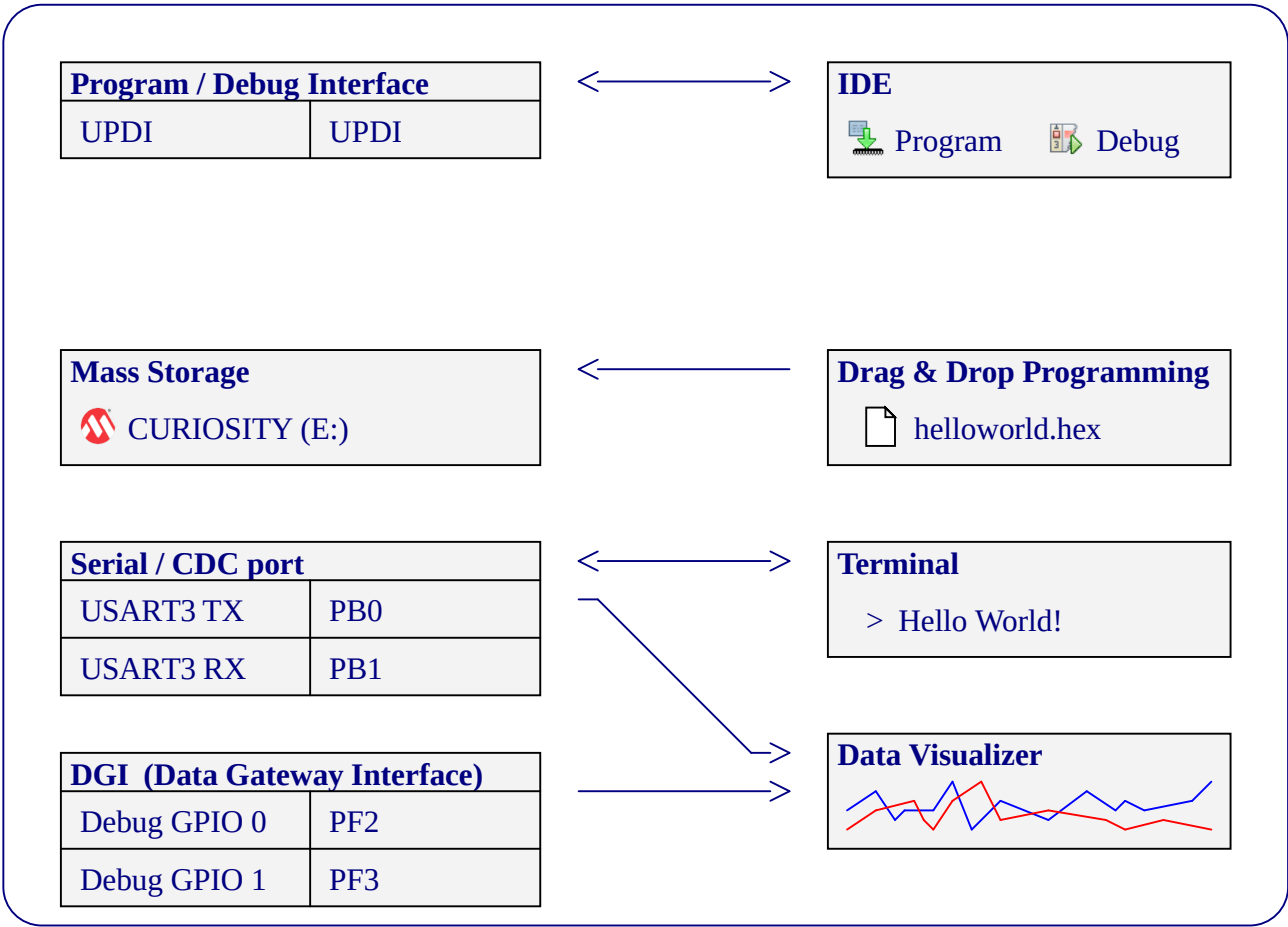


ATmega4809 Curiosity Nano



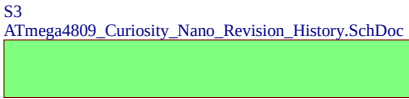
On-Board Peripherals

LED0	PF5	Active Low
SW0	PF6	Active Low



"Text in Silkscreen on PCB"

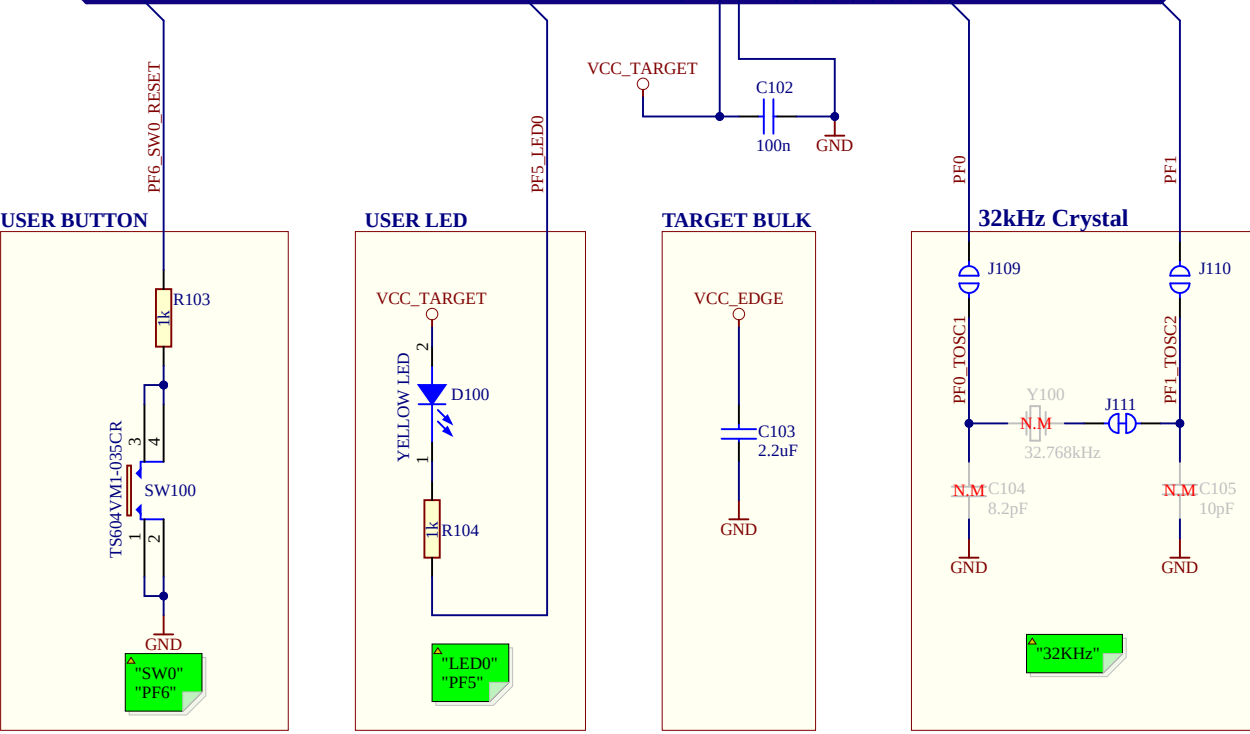
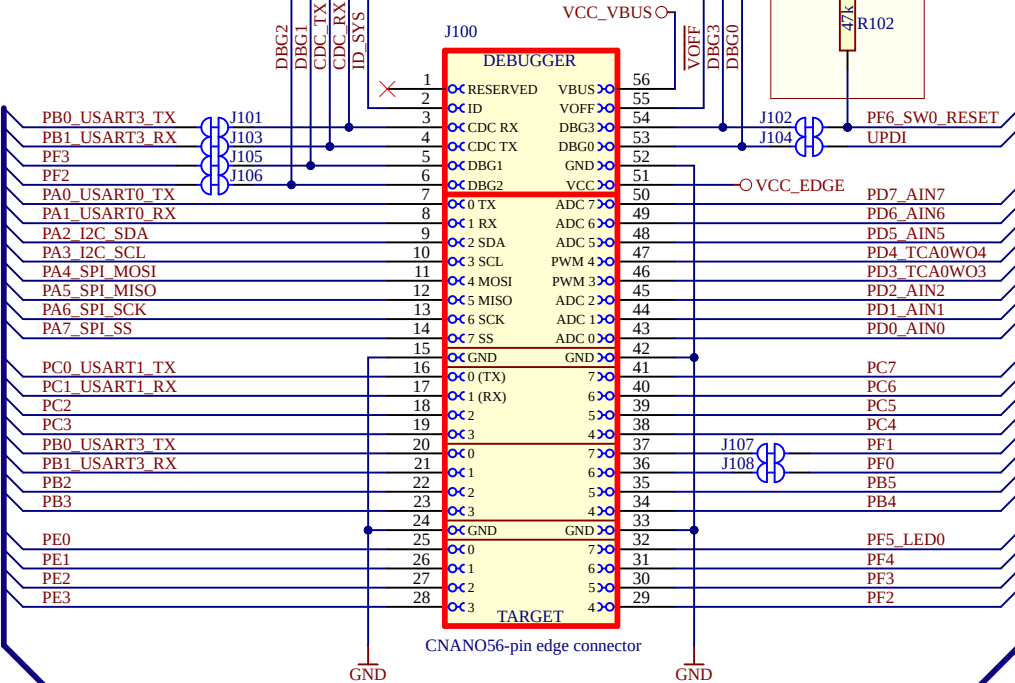
Project Owner: AH		 MICROCHIP				
PCB Layout Contact: TF, SLT						
PartNumber: DM320115		Project Title <i>ATmega4809 Curiosity Nano</i>		Variant: <u>Default Assembly</u>		
Sheet Title <u>Top Level</u>						 <i>Designed with</i> Altium Altium.com
Size	A3	SCH #: <u>A09-3094</u>	Rev: <u>9</u>	Date: <u>12.11.2024</u>		
		PCB #: <u>A08-2838</u>	Rev: <u>8</u>	Sheet <u>1</u> of <u>4</u>		
File: <u>ATmega4809_Curiosity_Nano_TopLevel.SchDoc</u>						



ATmega4809

	ATmega4809	
Debugger	Name	Pin
CDC TX	USART3 RX	PB1
CDC RX	USART3 TX	PB0
DBG0	UPDI	UPDI
DBG1	GPIO1	PF3
DBG2	GPIO0	PF2
DBG3	<u>RESET</u>	PF6
VTG	1.8V - 5.5V	

CDC_RX	CDC_RX
CDC_TX	CDC_TX
DBG0	DBG0
DBG3	DBG3
DBG1	DBG1
DBG2	DBG2
VOFF	VOFF
ID_SYS	ID_SYS



NOTE on Crystal calculations:

The calculations are based on a crystal used on an earlier revision of this board.

Crystal datasheet:
Ccrystal = 7pF
max ESR = 70kOhm
Accuracy ± 20 ppm

ATmega4809 datasheet:
Cxin = 5.5pF
Cxout = 5.5pF
 $Cl \approx 1/(1/(1/5.5pF) + (1/5.5pF)) \approx 2.75pF$
Maximum Load = 12.5pF
Maximum ESR = 80kOhm

Estimated $C_{pcb} = 0.5\text{pF}$

Estimated load
 $C = 2 (C_{\text{crystal}} - C_{\text{para}} - C_{\text{pcb}})$
 $C = 2 (7\text{pF} - 2.75\text{pF} - 0.5\text{pF})$
 $C = 7.5\text{pF}$

Selected in design after verification
C= 8.2/10pF

NOTE on UART/CDC:

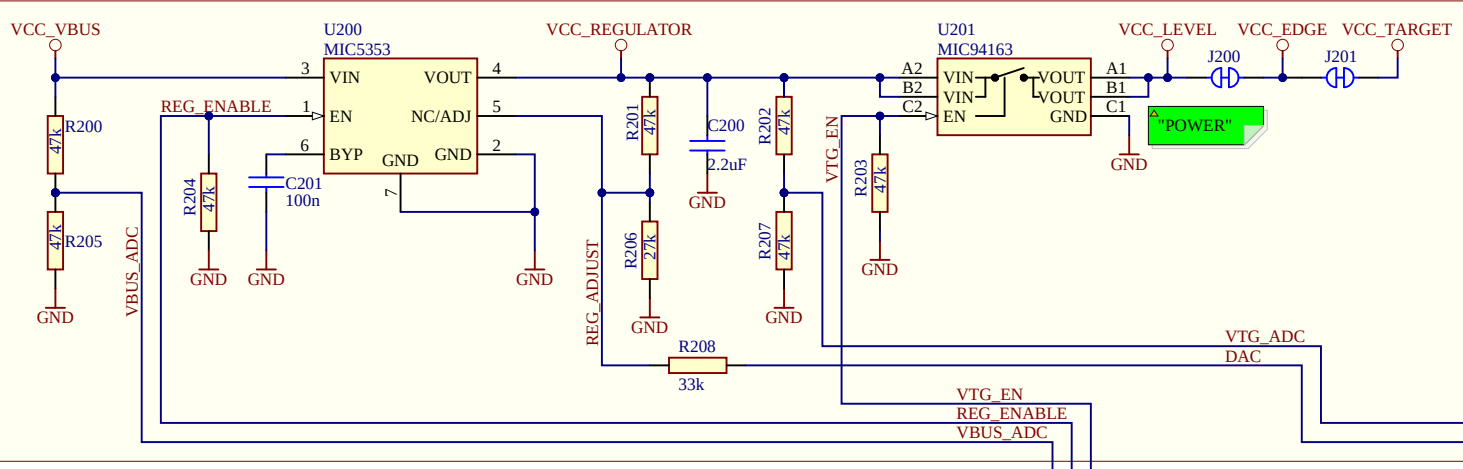
RX/TX on the header denotes the input/output direction of the signal respective to it's source.

CDC TX is output from the DEBUGGER.
CDC RX is input to the DEBUGGER.
TX is output from the TARGET device.
RX is input to the TARGET device.

NOTE on I2C:

No pull-ups on board. Pull-ups should be mounted close to client device(s).

TARGET ADJUSTABLE REGULATOR



J200:
- Cut-strap used for full separation of target power from the level shifters and on-board regulators.
- For current measurements using an external power supply, this strap could be cut for more accurate measurements. Leakage back through the switch is in the micro ampere range.
J201:
- For current measurements using the on-board power supply, this strap must be cut and an ammeter connected across.

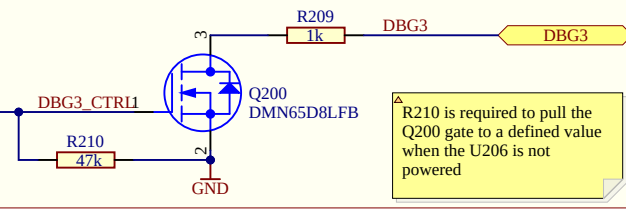
MIC5353:
Vin: 2.6V to 6V
Vout: 1.25V to 5.1V
Imax: 500mA
Dropout (typical): 50mV@150mA, 160mV @ 500mA
Accuracy: 2% initial
Thermal shutdown and current limit

Maximum output voltage is limited by the input voltage and the dropout voltage in the regulator.

Interface Signal	ICSPTM TARGET	UPDI TARGET	SWD TARGET
CDC TX	UART RX	UART RX	UART RX
CDC RX	UART TX	UART TX	UART TX
DBG0	DAT	UPDI	SWDAT
DBG1	CLK	GPIO	SWCLK
DBG2	GPIO	GPIO	SWO/GPIO
DBG3	MCLR	RESET	RESET
VCC	-	-	-

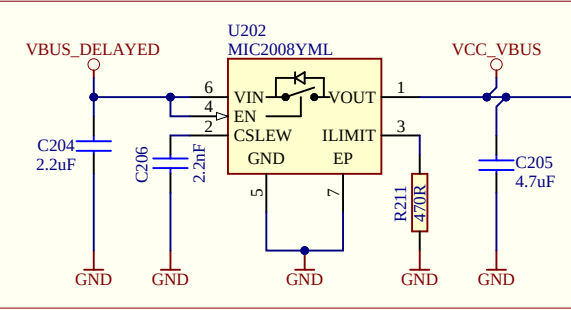
ADJUSTABLE OUTPUT AND LIMITATIONS:
- The DEBUGGER can adjust the output voltage of the regulator between 1.25V and 5.1V to the target.
- The voltage output is limited by the input (USB), which can vary between 4.40V to 5.25V
- The level shifters have a minimal voltage level of 1.65V and will limit the minimum operating voltage allowed for the target to still allow communication.
- The MIC94163 has a minimal voltage level of 1.70V and will limit the minimum voltage delivered to the target.
- Firmware configuration will limit the voltage range to be within the target specification.

DBG3 OPEN DRAIN



DEBUGGER

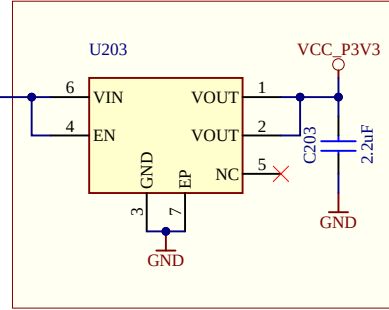
VBUS SLEW RATE- & CURRENT-LIMIT



SLEW RATE LIMIT:
- With C206 set to 2200pF, the slew rate of VCC_VBUS is limited to 2 V/ms by the power switch MIC2008.

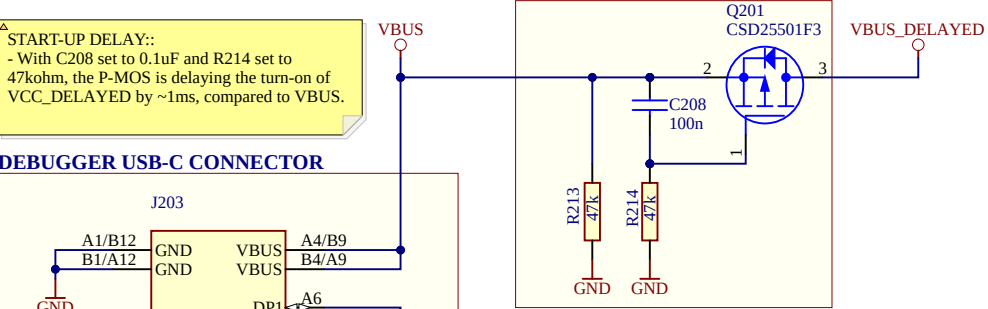
CURRENT LIMIT:
- With R211 set to 470ohm, the current through the power switch MIC2008 is limited to 500mA.

DEBUGGER REGULATOR

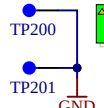
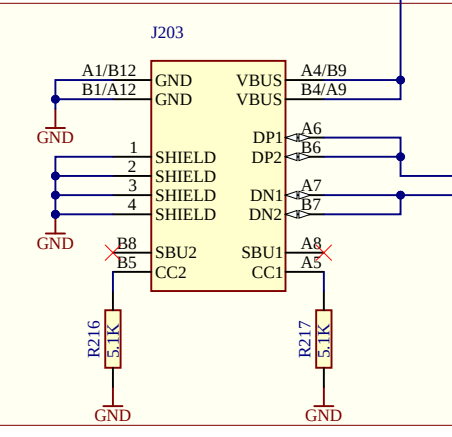


MIC5528:
Vin: 2.5V to 5.5V
Vout: Fixed 3.3V
Imax: 500mA
Dropout: 260mV @ 500mA

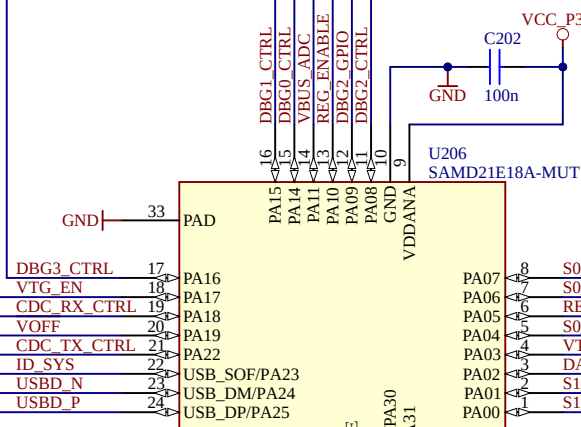
VBUS START-UP DELAY



DEBUGGER USB-C CONNECTOR



TP200 and TP201 are MTG Holes in the corners of the PCB in the DEBUGGER section labelled "GND" on the silkscreen.



Revision History

PCB Assembly Rev 1:

Design Changes:

Initial Design

PCB:

PCB revision 1

PCB Assembly Rev 2:

Design Changes:

Removed bypass switch

PCB:

PCB Revision 2, changed layout

PCB Assembly Rev 3:

Design Changes:

Added reverse blocking switch (U108)
Added ID_SYS pin
Added VÖFF pin
Change 10uf to 4.7uF (U100)

PCB:

PCB Revision 3
Increased board size for ID and VÖFF pin
Rerouted for target voltage blocking switch

PCB Assembly Rev 4:

Design Changes:

Added USB VBUS protection (U109)
Added VBUS pin

PCB:

PCB Revision 4
Increased board size for VBUS pin
Reroute the debugger section for the VBUS pin

PCB Assembly Rev 5:

Design Changes:

Added input cap on U109

PCB:

PCB Revision 5
Added input cap on U109

PCB Assembly Rev 6:

Design Changes:

Changed to Cut-Straps with thru-holes in Debugger Section.
Replaced Limit Switch with Fuse. Changed VBUS and VCC_VBUS nets back to previous scheme on DEBUGGER schematic. Added cut-straps to crystal

PCB:

PCB Revision 6
Changed layout to add Fuse and Cut-Straps with holes.
Adusted level-shift routing and adjusted Q101 footprint for improved assembly quality.

PCB Assembly Rev 7:

Design Changes:

Added solderstrap jumper with holes between edge connector power (VCC) and VCC_TARGET.

PCB:

PCB Revision 7
Updated J200 (edge pin header connector) with staggered footprint, and adjusted traks, polygons, teardrops, and text accordingly.

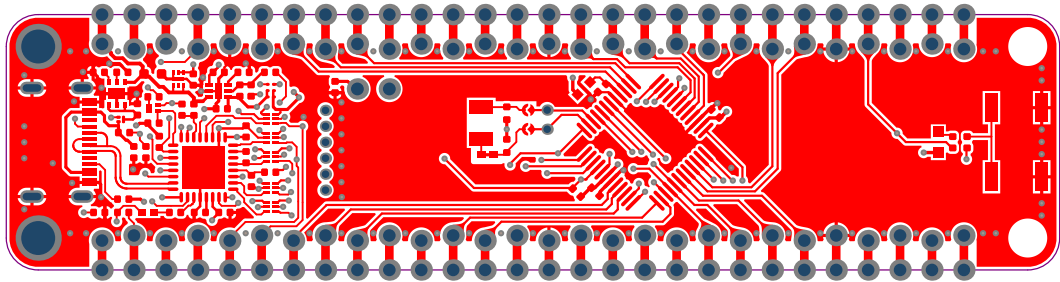
PCB Assembly Rev 9:

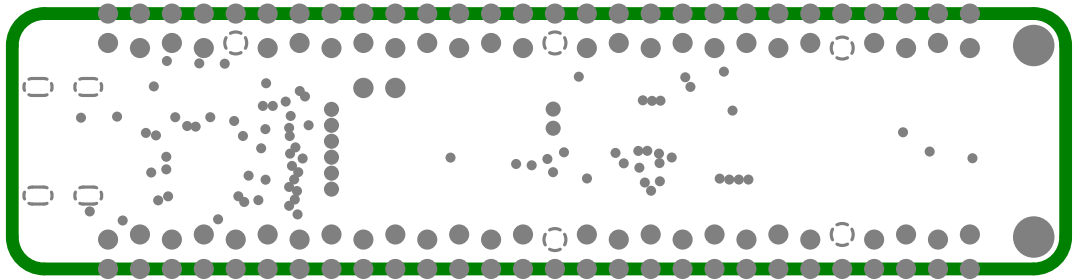
Design Changes:

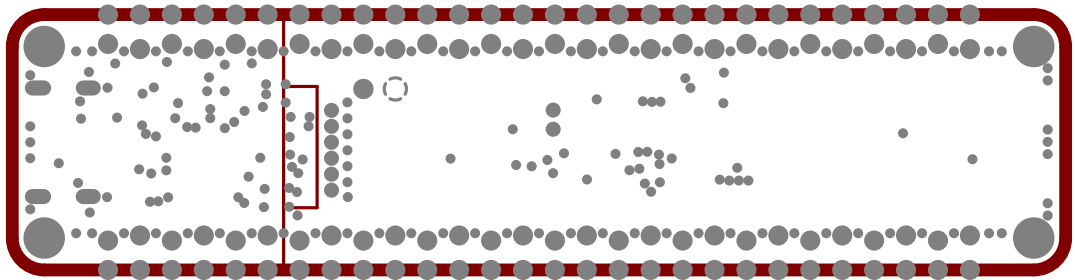
Designators reset
Power input and current limit redesign
USB-C Connector
32.768 kHz footprint disconnected from U100 by default
PF0 and PF1 connctected to edge conenctor by default

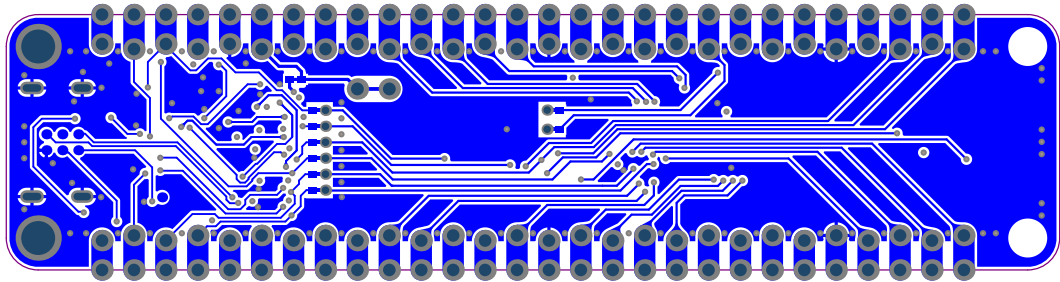
PCB:

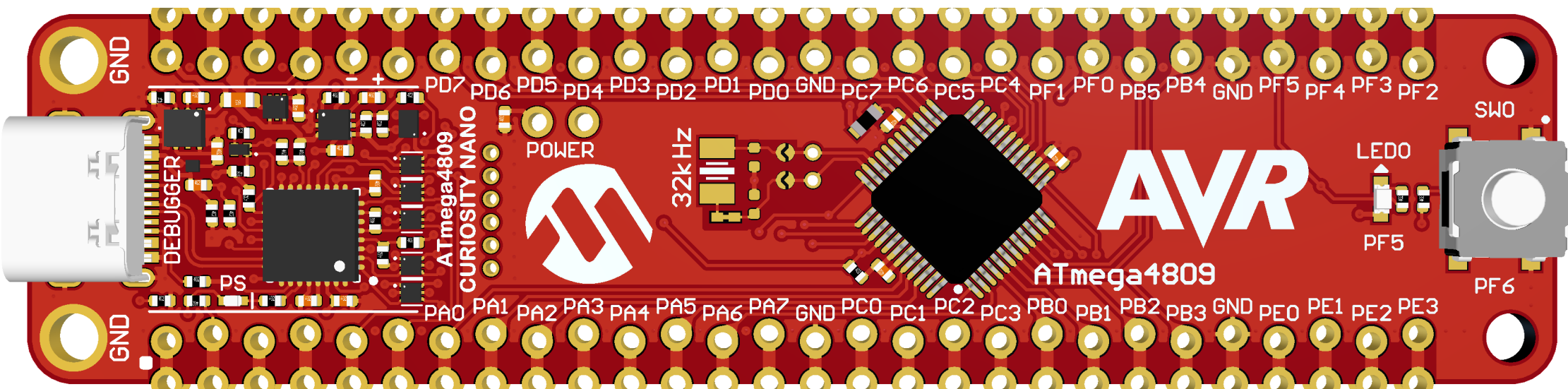
PCB Revision 8
Updated footprint for J100, U204, U205, U207, U208, U209
U100 placement changed slightly, and adjusted tracks, polygons, teardrops and text accordingly











Microchip © 2024

PCBA LABEL

A08-2838 Rev8

UK
CA

PF0
PF1



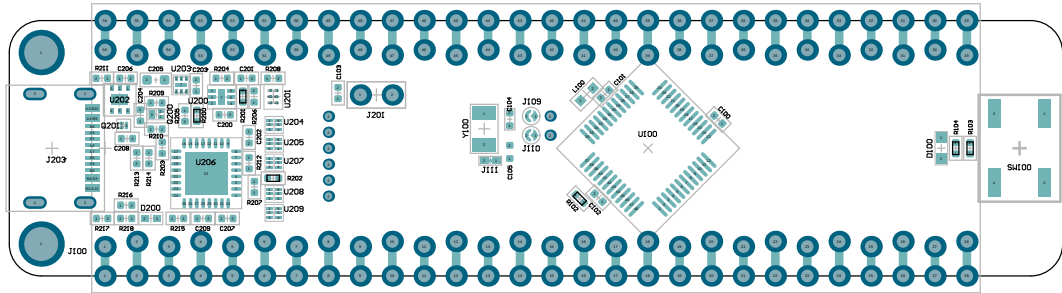
CONNECTIONS			
PF6	PF2	PF3	UPDI
PB0	PB1	D3	D2
		D1	D0
		RX	TX
		CDC	DEB
		BUGGER	

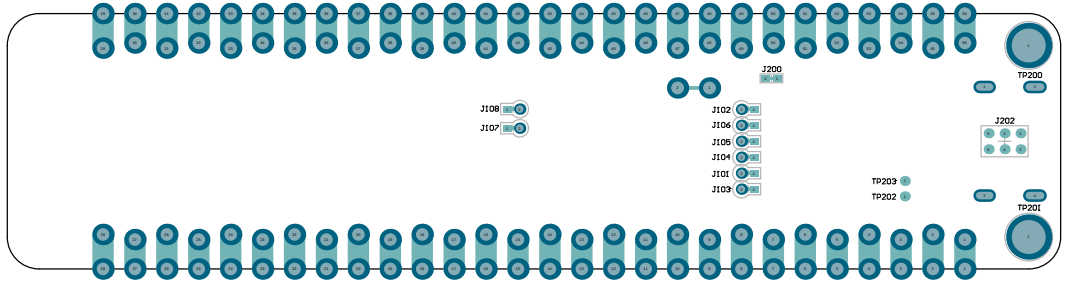


GND
BOOT

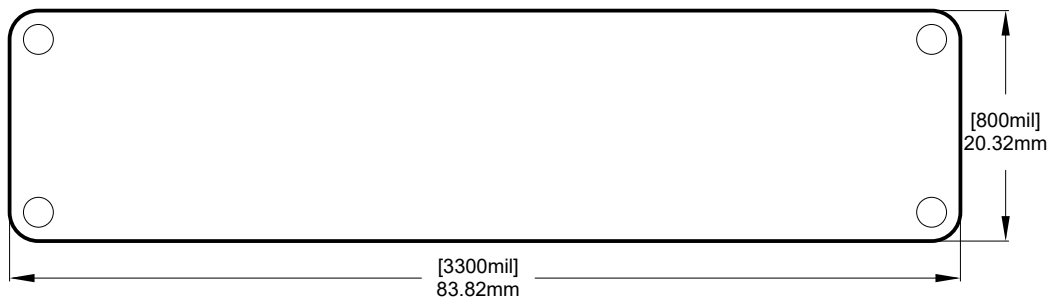
PF2 PF3 PF4 PF5 GND PB4 PB5 PF0 PF1 PC4 PC5 PC6 PC7 GND PD0 PD1 PD2 PD3 PD4 PD5 PD6 PD7 VTG GND D0 D3 VOFF VBUS GND

PE3 PE2 PE1 PE0 GND PB3 PB2 PB1 PB0 PC3 PC2 PC1 PC0 GND PA7 PA6 PA5 PA4 PA3 PA2 PA1 PA0 D2 D1 TX RX ID NC GND

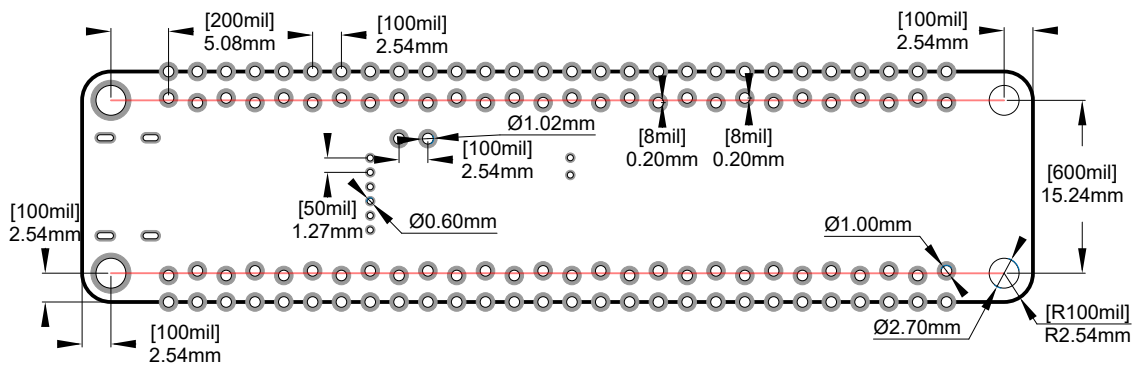




Board Outline

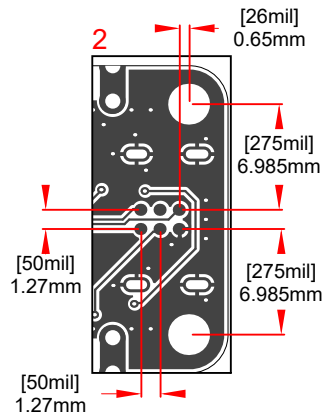
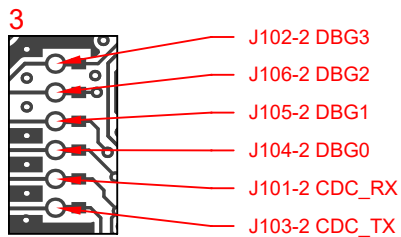
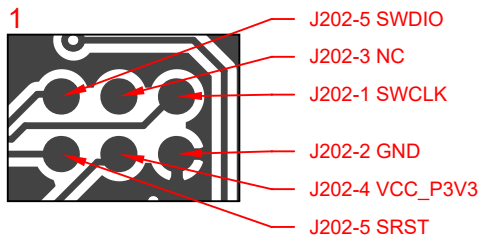
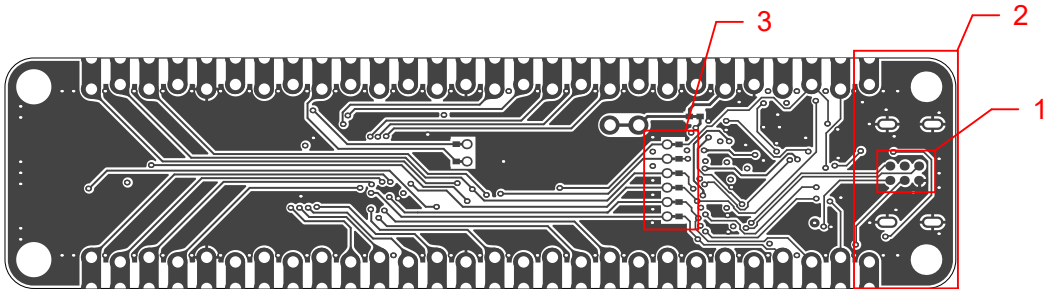
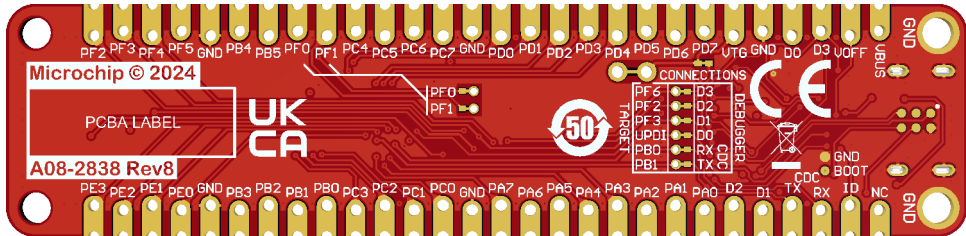


Through-Hole Dimensions



Centerline of staggered through-holes for pin headers.

Test Point Placement [Bottom View]



PCB Stackup

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.010mm	Solder Resist	Solder Mask	GTS
Copper	Top Layer	0.035mm		Signal	GTL
Prepreg		0.117mm	PP-016	Dielectric	
Copper	Gnd Plane	0.035mm		Internal Plane	GP1
Core		1.200mm	FR-4	Dielectric	
Copper	Pwr Plane	0.035mm		Internal Plane	GP2
Prepreg		0.117mm	PP-016	Dielectric	
Copper	Bottom Layer	0.035mm		Signal	GBL
Surface Material	Bottom Solder	0.010mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO
Total thickness: 1.594mm					

Component list

ATmega4809 Curiosity Nano

PCBA Number:

A09-3094

PCBA Revision:

9

Variant:

Default Assembly

Print Date: 12.11.2024 10:31:40



Fitted	Designator	Quantity	Value	Manufacturer	MPN	Description
Fitted	C100, C101, C102, C201, C202, C207, C208	7	100n	Yageo	CC0402KRX7R7BB104	Ceramic capacitor, SMD 0402, X7R, 16V, +/-10%
Fitted	C103, C200, C203, C204	4	2.2uF	TDK	C1005X5R1A225K050BC	CAP CER 2.2UF 10V 10% X5R 0402
Fitted	C205	1	4.7uF	Yageo	CC0603KRX5R6BB475	Ceramic capacitor, SMD 0603, X5R, 10V, 10% (de31036)
Fitted	C206	1	2.2nF	Kemet	C0402C222J3GACTU	Ceramic capacitor, SMD 0402, C0G, 25V, +/-5%
Fitted	C209	1	1u	Kemet	C0402C105K9PAC	Ceramic capacitor, SMD 0402, X5R, 6.3V, +/-10% (de26942)
Fitted	D100	1	YELLOW LED	ROHM	SML-D12Y1WT86	LED, SMD 0603, Yellow, Wave length=590nm, 100mcd @ (20mA, 2.2Vf) rohm
Fitted	D200	1	GREEN LED	ROHM	SML-P12MTT86R	LED, SMD 0402, Green, Wave length=569nm, 2.1mcd @ (1mA, 1.9Vf) rohm
Fitted	FW1	1	nEDBG firmw are			nEDBG firmw are
Fitted	J203	1	L-KLS1-5409-R	Cabcon A/S	L-KLS1-5409-R	USB2.0 Type-C, Surface mount signals and DIP shield
Fitted	L100	1	BLM18PG471SN1D	Murata	BLM18PG471SN1D	SMD RF inductor 0603. Z=470Ohm (@100MHz), Max R(dc)=0.20Ohm, Max current=1A
Fitted	LABEL1	1	Label PCBA	ACT Logimark AS	505462	PCBA identification label PP Top White Gloss
Fitted	PCB1	1	A Tmega4809 Curiosity Nano PCB documentation			A Tmega4809 Curiosity Nano PCB documentation
Fitted	PCBADOC1	1	A09-3094 PCBA files			A Tmega4809 Curiosity Nano PCBA documentation
Fitted	Q200	1	DMN65D8LFB	Diodes Incorporated	DMN65D8LFB-7	N-channel MOSFET, DFN1006-3 (SOT883), 60V, 330mA, 40hm
Fitted	Q201	1	CSD25501F3	Texas Instruments	CSD25501F3	P-Channel MOSFET, LGA, 20V, 3.6A, 76mOhm
Fitted	R102, R200, R201, R202, R203, R204, R205, R207, R210, R213, R214	11	47k	KOA	RK73H1ETTP4702F	Thick film resistor, SMD 0402, 1/16W, 1%
Fitted	R103, R104, R209, R212, R215, R218	6	1k	Yageo	RC0402FR071KL	Thick film resistor, SMD 0402, 1/16W, 1%
Fitted	R206	1	27k	Yageo	RC0402FR-0727KL	Thick film resistor, SMD 0402, 1/16W, 1%
Fitted	R208	1	33k	Yageo	RC0402FR-0733KL	Thick film resistor, SMD 0402, 1/16W, 1%
Fitted	R211	1	470R	Yageo	RC0402FR-07470RL	Thick film resistor, SMD 0402, 1/16W, 1%
Fitted	R216, R217	2	5.1K	Panasonic	ERJ-2RK5F5101X	Thick film resistor, SMD 0402, 1W, 1%
Fitted	SW100	1	TS604VM1-035CR	Dailywell Electronics Co.LTD	TS604VM1-035CR-R	SWITCH, SMD, 260gf, 6.4mm X 6.2mm
Fitted	TEST1	1	A Tmega4809 Curiosity Nano test			Fixture test for A Tmega4809 Curiosity Nano
Fitted	TESTDOC1	1	Curiosity Nano Test Instructions			Generic Test Instructions for Curiosity Nano
Fitted	U100	1	A Tmega4809-AFR	Microchip	A Tmega4809-AFR	Atmel 8-bit RISC MCU, 48K Flash, 48pin TQFP, 7x7mm, 0.5mm pitch package
Fitted	U200	1	MIC5353	Microchip	MIC5353YMT-TR	500mA Ultra Low Dropout LDO regulator, 2% accuracy, 1.6x1.6mm MLF
Fitted	U201	1	MIC94163	Microchip	MIC94163YCS-TR	Loadsw itch, Rds(on) = 14.5mohm, 1.0mm x 1.5mm WLCSP, reverse blocking
Fitted	U202	1	MIC2008YML	Microchip	MIC2008YML-TR	0.2A- 2.5A adjustable current limit power sw itch, 2mm x 2mm DFN
Fitted	U203	1	MIC5528-3.3YMT	Microchip	MIC5528-3.3YMT-T5	LDO 3.3V 0.5A 6TDFN
Fitted	U204, U205, U207, U208, U209	5	74LVC1T45FW4-7	Diodes Incorporated	74LVC1T45FW4-7	Single-Bit Dual-Supply Transceiver, 1.65-5.5 Translation and 3-State Outputs
Fitted	U206	1	SAMD21E18A-MUT	Microchip	ATSAMD21E18A-MUT	32-bit RISC MCU 32pin
Not Fitted	C104	0	8.2pF	Murata	GCQ1555C1H8R2D	8.2pF GCQ15 Series HiQ Automotive 0402(EA)
Not Fitted	C105	0	10pF	Yageo	CC0402JRNPO9BN100	Ceramic capacitor, SMD 0402, NP0, 50V, +/-5% (de16674)
Not Fitted	Y100	0	32.768kHz	Microchip	VMK3-9001-32K7680000TR	Crystal, 32.768kHz, CL=9.0pF, ESR=70kOhm, SMD LxW=3.2 x 1.5mm, 20ppm

Approved

Notes